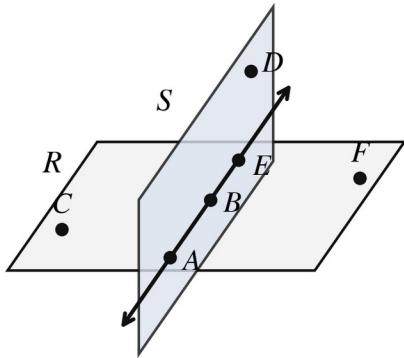


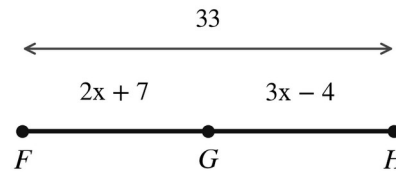
Every skill on the real test is here — the questions are just asked a different way. Show all your work. Circle your final answer.

Use the diagram for question 1.



1. Planes R and S intersect.
 - a) Name the intersection of the two planes.
 - b) Name three collinear points.
 - c) Name a point that is NOT coplanar with A, B, and C.
2. Sketch and label: line m intersecting plane W at exactly one point, T .
3. True or false? Any three points are collinear. Explain, and sketch a counterexample if it is false.

Use the figure for question 4.

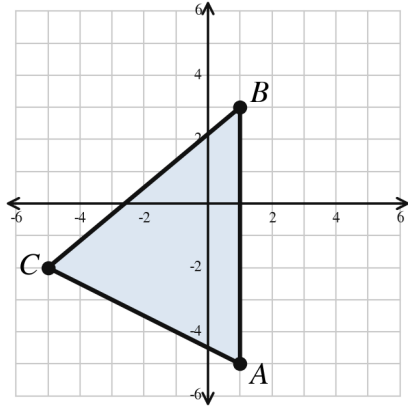


4. Find x . Then find FG and GH .
5. A straight trail runs 6.8 miles from the parking lot to the summit. A lookout is on the trail, 2.9 miles from the parking lot.
 - a) How far is the lookout from the summit?
 - b) A ranger station sits on the trail at the midpoint between the lookout and the summit. How far is it from the parking lot?
6. $M(-1, 3)$ is the midpoint of $\overline{AB}$, and A has coordinates $(-6, -1)$. Find B . Then find the length AB .
7. Is $\triangle PQR$ isosceles? $P(-2, 1)$, $Q(4, 1)$, $R(1, -3)$. Use the Distance Formula and explain.

Honors Geometry – Chapter 1 Practice Test
Page 2

8. Sketch a concave hexagon. Then name a polygon with 10 sides.

9. Use $\triangle ABC$ below. Each square is 1 unit.

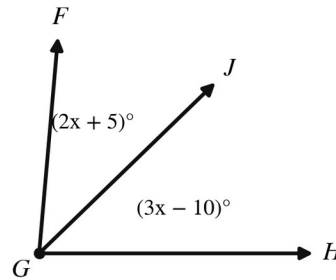


- Find the perimeter of $\triangle ABC$. Round to the nearest tenth.
 - Find the area of $\triangle ABC$. (Which side is the base? Where is the height?)
10. Without a protractor, describe how to construct a 22.5° angle starting from a straight angle. How do you KNOW it is exactly 22.5° ?
11. $\angle A$ and $\angle B$ are complementary. $m\angle A = (5x - 3)^\circ$ and $m\angle B = (2x + 9)^\circ$. Find both measures.

Full Name: _____

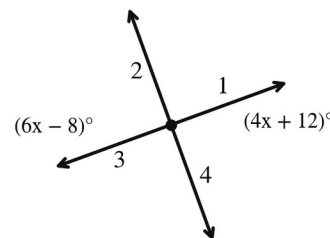
Period: _____

Use the figure for question 12.



12. $m\angle FGH = 85^\circ$. Find x , then find $m\angle FGJ$ and $m\angle JGH$.
13. Ray QS bisects $\angle PQR$. $m\angle PQS = (3x + 9)^\circ$ and $m\angle PQR = (8x - 6)^\circ$. Find $m\angle PQR$.

Use the figure for question 14.



14. $m\angle 2 = (6x - 8)^\circ$ and $m\angle 4 = (4x + 12)^\circ$.
- Name a pair of vertical angles and a linear pair.
 - Find x . c) Find $m\angle 1$.
15. $\angle C$ and $\angle D$ are supplementary. $m\angle C = (7x + 2)^\circ$ and $m\angle D = (3x - 12)^\circ$.
- Find both measures.
 - Must $\angle C$ and $\angle D$ form a linear pair? Explain.